

Installation Guide

Complete step-by-step guide to deploying OQUL EdTech OS from source to production. Estimated time: **30–60 minutes**.



Prerequisites Checklist

Before starting, ensure you have: Node.js 20+, Git, Docker Desktop, a Groq API key (free at console.groq.com), and PostgreSQL credentials.

Phase 1 — Environment Setup

Clone the Repository

1

```
# Clone OQUL to your local machine
git clone https://github.com/your-org/oqul-platform.git
cd oqul-platform
```

Install Dependencies

2

```
# Install all Node.js packages
npm install
```

Configure Environment Variables

3

```
# Copy the example environment file
cp .env.example .env.local

# Edit with your values
nano .env.local
```

Environment Variables Reference

Variable	Description	Required
DATABASE_URL	PostgreSQL connection string	Required
REDIS_URL	Redis connection string	Required
GROQ_API_KEY	Groq AI API key for Leila tutor	Required
JWT_SECRET	Secret for JWT token signing (32+ chars)	Required

Variable	Description	Required
NEXTAUTH_URL	Your application URL	Required
SMTP_HOST	Email server hostname	Optional
SMTP_USER	Email server username	Optional
SMTP_PASS	Email server password	Optional
SENTRY_DSN	Sentry error tracking DSN	Optional

Phase 2 — Database Setup

Start PostgreSQL and Redis via Docker

1

```
# Launch all required services
docker compose -f docker-compose.runtime.yml up -d

# Verify services are running
docker compose ps
```

Run Database Migrations

2

```
# Apply all schema migrations
npx drizzle-kit migrate

# Expected: 17 migrations applied
```

Seed the Curriculum Data

3

```
# Seed full K-12 curriculum (takes ~5 minutes)
node scripts/seed-full-curriculum.mjs

# Seed primary school specifically
node scripts/seed-phase52-primary.mjs

# Seed secondary school (2BAC/1BAC/TC)
node scripts/seed-phase53-full-secondary-school.mjs
```

Phase 3 — Launch Application

Start Development Server

1

```
npm run dev
```

Verify Installation

- 2
- Open your browser and navigate to `http://localhost:3000`. You should see the OQUL landing page. Register a student account and verify the AI tutor Leila responds.

Run Test Suite

- 3
- ```
Run all unit and integration tests
npm run test

Expected: All tests pass
```

Production Deployment Checklist

- ☐ All environment variables set in production `.env`
- ☐ `DATABASE_URL` points to production PostgreSQL instance
- ☐ `REDIS_URL` points to production Redis (with Sentinel for HA)
- ☐ `GROQ_API_KEY` validated and rate limits understood
- ☐ `JWT_SECRET` is cryptographically random (32+ characters)
- ☐ `NEXTAUTH_URL` set to your production domain (`https://`)
- ☐ All database migrations applied: `npx drizzle-kit migrate`
- ☐ Full curriculum seeded via seed scripts
- ☐ Prometheus + Grafana monitoring configured
- ☐ Sentry DSN configured for error tracking
- ☐ Email SMTP service configured for user verification
- ☐ SSL/TLS certificate configured (via nginx or Caddy)
- ☐ Load tests run: `k6 run load-tests/ai-runtime-k6.js`
- ☐ GitHub Actions CI pipeline green



Need help? Check [FAQ.pdf](#) first

The most common installation issues are covered in [05\\_SUPPORT/FAQ.pdf](#). If your issue is not covered there, consult [05\\_SUPPORT/Support\\_Guide.pdf](#) for how to get assistance.

Architecture Quick Reference

Port Reference

|      |                     |
|------|---------------------|
| 3000 | Next.js Application |
| 5432 | PostgreSQL          |
| 6379 | Redis               |

Key Directories

|                          |                        |
|--------------------------|------------------------|
| <code>src/app/</code>    | Next.js pages & routes |
| <code>src/server/</code> | Business logic         |

|      |            |
|------|------------|
| 9090 | Prometheus |
|------|------------|

|      |         |
|------|---------|
| 3001 | Grafana |
|------|---------|

|               |                 |
|---------------|-----------------|
| src/features/ | Feature modules |
|---------------|-----------------|

|                      |                 |
|----------------------|-----------------|
| curriculum-registry/ | Curriculum JSON |
|----------------------|-----------------|

|          |                    |
|----------|--------------------|
| scripts/ | Seed & setup tools |
|----------|--------------------|